

Banff National Park Activity Trends

Activity Trends 2020-25

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1 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on Treaty 8 and Douglas Treaties Territories as well as the traditional and unceded territories of First Nations and Métis Peoples across all regions of British Columbia, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). The program is administered by the US Geological Survey (USGS) and implemented by the North by Northwest Bat (NNW) Hub in British Columbia, Alberta and S.E Alaska.

NNW Bat Hub was established in 2024 with collaboration from Government of British Columbia, Government of Alberta, and Government of Alaska. The hub was formed by combining the previous Alberta Hub and the BC and southeast Alaska Hub. The implementation of the program within Alberta has key support from Parks Canada, government of Alberta staff, other wildlife biologists, Indigenous Nations, community members and naturalists across the province.

2.2 Threats to bat populations in Alberta

Bat populations in Alberta face a wide range of threats including forestry practices, anthropogenic expansion, and climate change among others. Of these threats, White-Nose Syndrome and wind energy development have emerged as two of the highest-priority concerns.

2.2.1 White-Nose Syndrome

White-nose Syndrome (WNS) is a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (*Pd*) that affects hibernating bats. As of 2026, WNS has been confirmed in nine Canadian provinces including Alberta, and *Pd* has been detected within British Columbia at two locations (Figure 1: (cwhc-rCSF2024Canadian?)). Continued monitoring of populations in AB is critical as *Pd* and WNS continue to spread throughout the province.

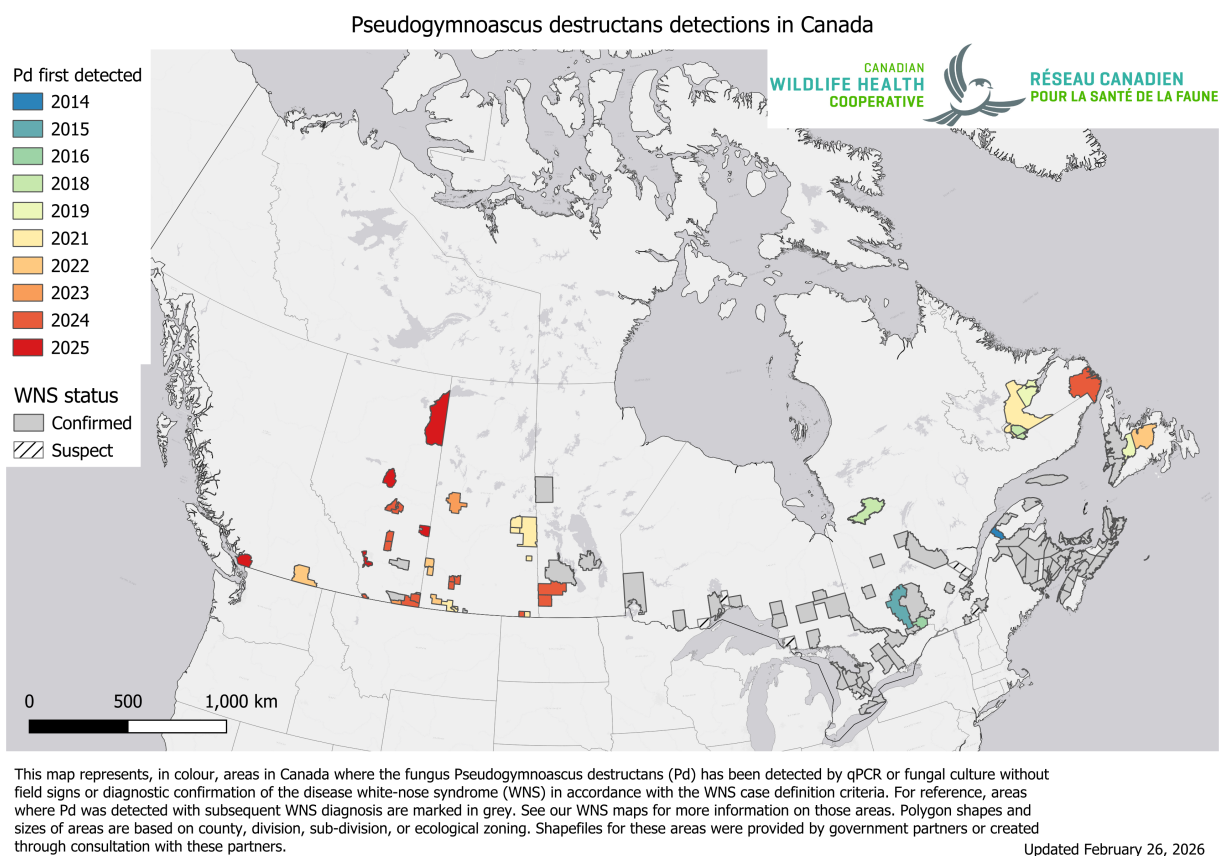


Figure 1: White-nose syndrome (WNS) and *Pseudogymnoascus destructans* (Pd) occurrence in Canada. Source: CWHC-RCSF (2026)

2.2.2 Wind Energy Development

Wind energy farms has been linked to high mortality rates across all bat species in Canada, with long distance migratory species accounting for 73% of recorded fatalities ([zimmerling2016Bat?](#)). In British Columbia alone, one estimate predicts an annual mortality rate of 680 bats per wind turbine ([zimmerling2016Bat?](#)). There are currently close to 300 turbines active in the province across 10 different projects with the most turbines found in the Peace region (Figure 2). In late 2024, B.C announced nine new wind projects which would require the installation of over 300 new wind turbines across the province ([ministryofenergyandclimatesolutions2024BC?](#)). As of 2023, COSEWIC designated all three of B.C.'s long distance migratory species as endangered due in part to fatalities at wind turbine facilities ([committeeonthestatusofendangeredwildlifeincanada2023Hoary?](#)).



Figure 2: Current wind turbines in Alberta based on the Canadian Wind Turbine Database (Source: <https://search.open.canada.ca/openmap/79fdad93-9025-49ad-ba16-c26d718cc070>)

2.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data:

1. **Imperfect species identification** — Unlike birds, bats rely on echolocation rather than song, and species with similar ecology and physiology often produce overlapping calls. While detectors are deployed to maximize recordings of diagnostic Search Phase calls, species identifiability will always vary depending on the local assemblage.
2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, while smaller species with quieter calls or specialized habitat use are less reliably recorded. As a result, trend estimates cannot be produced for all species using acoustic methods alone.
3. **Data variability** — Many factors influence recording quality and bat activity, including wind speed, humidity, and local habitat. Additionally, stationary detectors cannot account for individual recapture, meaning acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets but has known high misclassification rates for bats, making manual verification necessary to ensure accurate results.

3 Methods

Banff National Park has monitored one NABat grid cell using three sites and has collected acoustic data from 2020 to 2025. Recordings were processed using an automated classifier (Kaleidoscope’s Bats of North America 5.4.0 classifier) to assign species identifications and exclude non-bat recordings.

Manual verification was only performed in 2024 and 2025. For the trends estimates only autoID results were included unless the manual verification results indicated that what was tagged as a bat was actually noise. We only included the autoID results for the trends to keep it consistent, since looking at the manually verified results for only the last two years would have made it difficult to interpret the results across the years.

3.1 Manual Verification of 2025 Data

3.2 Data processing

Species presence and nightly activity were calculated from a site-night-species table constructed for all sampled nights. Sampling effort was defined as each unique site-night with acoustic recordings. Species codes were standardized across data sources prior to analysis. Weather covariates were summarized for each sampling night and included mean temperature, mean relative humidity, total precipitation, and mean wind speed. Moon covariates were calculated for each site-night and included moon illumination, and an indicator of moon position. Mean nightly passes included all recordings with species or couplet labels, with couplets split and counted for both species. For trend analyses, MYLU, MYSE, and MYVO were pooled into a single *Myotis* group (40KMYO) - because

we know that there is high degree of error in the autoID for these species. That being said most of the recordings are likely to be MYLU.

3.3 Statistical analysis

Statistical trend analyses were conducted for EPFU, LABO, LACI, LANO, and 40KMYO. Models were fit to nightly detection counts using negative binomial generalized linear mixed models (GLMMs) with a log link. Sites were included as a random intercept. Candidate model structures included year, Julian date, quadratic Julian date, and species-specific covariates. Trend analyses are presented only for species confirmed to be present at the sites through manual verification and for which models fully converged.

4 Results

Across the 2020–2025 monitoring period, a total of 35,103 bat detections were recorded across the three sites in the Banff NABat grid cell. The Myotis group (40KMYO) had the highest number of detections (29,209; mean = 111.00). This was followed by Hoary Bat (2,009; mean = 7.61), Silver-haired Bat (1,773; mean = 6.72), Big Brown Bat (1,355; mean = 5.13), and Eastern Red Bat (757; mean = 2.87).

4.1 Species detected and mean nightly activity

Detected species and nightly bat activity varied among years and sites (Table 1). In total, up to eight species were detected, with Big Brown Bat (EPFU), Silver-haired Bat (LANO), Hoary Bat (LACI), and Eastern Red Bat (LABO) detected across most years and sites. Myotis species showed more variable detections: Little Brown Myotis (MYLU) was detected frequently, while Long-legged Myotis (MYEV), Western Long-eared Myotis (MYVO), and Long-eared Myotis (MYSE) were detected less consistently.

Nightly bat activity varied across quadrants and years. Activity was highest in NE and NW in 2022–2023, while SW remained consistently low. NW showed a pronounced peak in 2024, the highest across all years, whereas activity in 2025 was more uniform among quadrants. Overall, while several species were consistently detected across the study period, both species presence and nightly activity showed strong spatial and temporal variability. Differences in sampling effort among quadrants and years likely contributed to some level to this pattern.

4.2 Species Trends

Models converged for five species/groups: Big Brown Bat (EPFU), Eastern Red Bat (LABO), Hoary Bat (LACI), Silver-haired Bat (LANO), and the combined Myotis group (@fig-trends). Significant increases in nightly activity were detected for Big Brown Bat ($\beta = 0.74$, SE = 0.09, $z = 8.11$, $p < 0.001$) and Eastern Red Bat ($\beta = 0.81$, SE = 0.10, $z = 7.88$, $p < 0.001$). In contrast, no significant temporal trend was detected for Hoary Bat ($\beta = 0.10$, SE = 0.09, $z = 1.10$, $p = 0.270$), Silver-haired Bat ($\beta = 0.06$, SE = 0.13, $z = 0.48$, $p = 0.633$), or the combined Myotis group ($\beta = 0.22$, SE = 0.12, $z = 1.81$, $p = 0.071$).

Banff bat trends

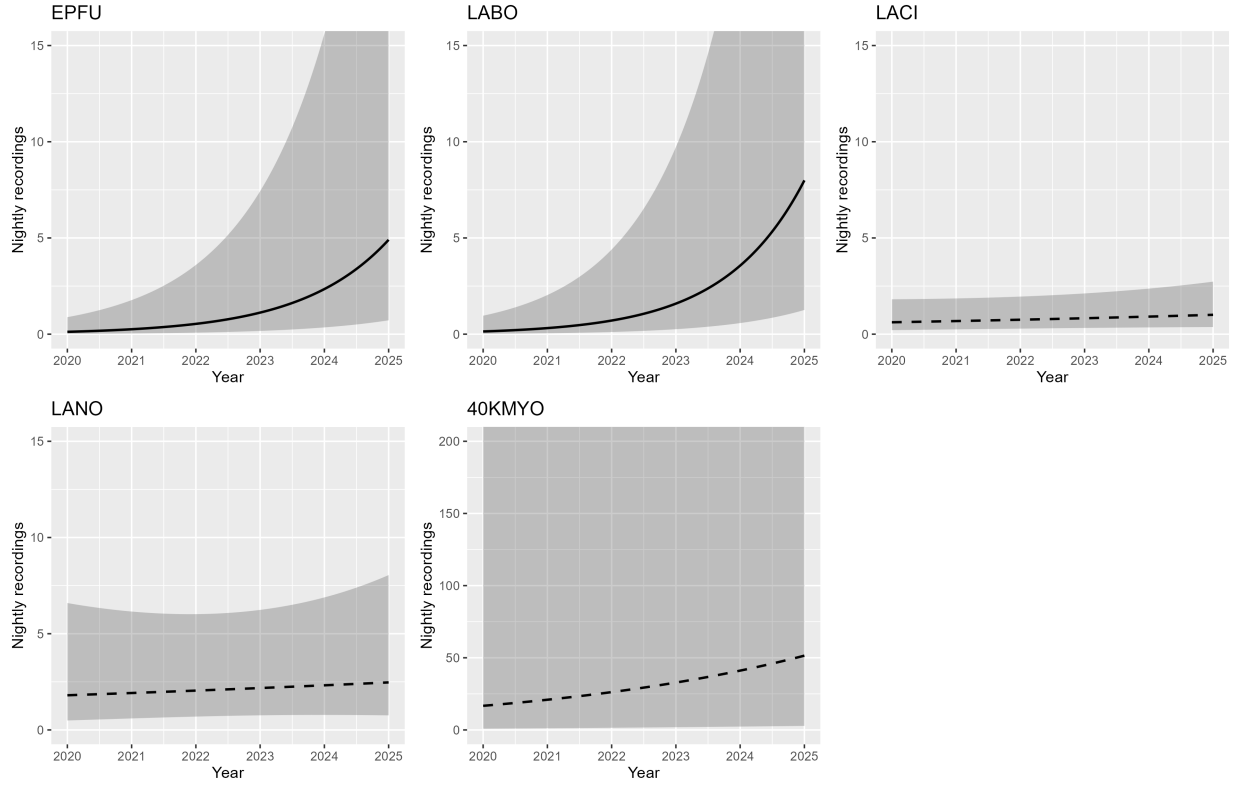


Figure 3: Species relative activity trends from 2020–2025 in the Banff NABat grid cell. Lines show trend estimates of nightly detections with 95% confidence intervals. Solid lines indicate significant trends ($p < 0.05$), and dashed lines indicate non-significant trends ($p > 0.05$). Models are based on nightly acoustic detections and fitted using negative binomial mixed-effects models.

5 Discussion

Data used for trends only uses autoID from Kaleidoscope, which has been shown to confuse *Myotis* spp. for LABO.

Results must be taken with caution because of small sample size, but overall can conclude that bat species at the grid cell are stable.

6 Acknowledgements

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